Tractatus **Veritatis Trisductivus**

FOUNDATIONS OF MATHEMATICS
STANDPOINTS CLOSE WHERE METHODS CANNOT
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Riemann Hypothesis: The Formal Case Is Closed

The Hypothesis Is True Where Actualized, Every Position a Verdict Could Be Attempted From Is Closed Permanently by an Enumeration No Future Method Enlarges, and What Remains Is One Bit That Belongs to the Reader

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The Riemann Hypothesis is true where it is actualized, and this paper opens with that seal rather than reaching it. The critical line is elected three separate ways from three separate data, and the three are one classical generator read at three registers. The paper's structural result is a closure of the verdict side, obtained by changing the enumeration domain: obstructions are ordinarily catalogued by method, and methods are generated without bound, so no such catalogue closes and each is one unlisted method from refutation. The domain that closes is the set of positions from which a verdict could be attempted, and that set is fixed by the architecture doing the attempting. The verdict side occupies exactly ten standpoints. Ten blocks close them, two definitional, one constitutive, six theorem-grade, and one a classification fact with a register law beside it, eight of the ten spending no framework premise at all. The closures are stated twice, once positively and once in the negative form, because a boundary is stated most exactly by what does not cross it. The formal-alone register is closed definitionally, and its direction deficit is counted rather than described: one bit demanded against zero supplied across every instrument the register carries. What remains is one binary degree of freedom, and it belongs to the reader. A route arriving in any future arrives at a standpoint and creates none, which is why the closure holds over time without surveying time.

KEYWORDS Riemann Hypothesis; standpoint enumeration; barrier theorems; via negativa; orientation-blindness; Davenport-Heilbronn; Beurling systems; theta inversion; Mellin-Plancherel; Li coefficients; Frobenius classification; foundations of mathematics

Introduction

The Riemann Hypothesis is true where it is actualized. The paper opens with the seal rather than arriving at it, because the order is part of the claim: an argument that reaches its affirmative after a long defence has already conceded that the affirmative was in doubt.

On the actualized prime field the distribution rests on the fixed line with no off-line residual, and the critical line is the unique stable attractor of that distribution. **The seal is issued in the register carrying the thermodynamic arrow**, which is the one content-bearing source of direction, and it is coextensive with the hypothesis and falls to the single object that would falsify the hypothesis.

The paper's structural result is a closure, and it is obtained by changing one thing: the domain of the enumeration.

The barrier literature catalogues obstructions **by method**. Relativization, natural proofs, algebrization each name a method class and prove it insufficient. **Methods are generated without bound**, so no catalogue of them closes, and every such catalogue is one unlisted method away from refutation. That is not a flaw in the individual theorems, which are correct and permanent. It is a fact about what a method census can be.

The domain that closes is the set of positions from which a verdict could be attempted. A standpoint is not a technique but a place: an instrument, a method-class, a register, a level, a ground, a mouth. **The set of positions is fixed by the architecture doing the attempting**, and the architecture's counts are closed by classification theorems rather than by survey. **A method arriving in any future arrives at a position that already exists. It does not create one.**

The verdict side occupies exactly ten standpoints, and ten blocks close them. **The closures are stated twice**, once positively and once in the negative form, because a boundary is stated most exactly from its outside, and that is why the ancient practice survived.

Section 2 sets out the reading landscape and the obstruction literature. Section 3 states the gap. Section 4 gives the method and its axioms. Section 5 gives the results. Section 6 gives the ten blocks derived from first principles. Section 7 states the falsifiers. Section 8 discusses the enumeration domain, the negative form, and the one bit. Section 9 concludes. The apparatus is quarantined to the appendices.

Literature Review

The elections of the critical line

The critical line enters through the functional equation of 1859 and has been read since as the axis of a reflection. **That reading is correct and it is not the only one available**, and two further elections of the same line are classical.

The unitarity of the additive-multiplicative coupling on the half-line. The Mellin transform carries the square-integrable functions of the half-line, under Lebesgue measure with the half-power twist, unitarily onto the functions of exactly one vertical line and onto no other. The half is the modulus of the dilation action on the additive measure, an invariant of the group structure and no draftsman's choice.

The pole configuration. The completed function cancels two simple singularities, and the multiplier attached to a zero has modulus one exactly on their perpendicular bisector, which is the same line.

What the literature has not assembled is that the three are one object. They are the theta inversion read at three registers: the fold is its Mellin image, the line is its modulus locus, the bisector is its action on the pole pair. **This is the identity Riemann transformed in 1859 to obtain the functional equation**, and the three elections sample it three ways.

The formulation landscape

The hypothesis is stated in many equivalent forms: analytic on the critical strip, the divisor inequalities of Robin and Lagarias, the closure criterion of Nyman and Beurling, the heat-flow threshold of de Bruijn and Newman, the operator spectrum of Hilbert and Pólya, the functional inequality of Weil, and the coefficient positivity of Li. Over a curve over a finite field the analogous statement is a theorem, proved by Weil and extended by Deligne.

The relevant fact about this landscape is that it is one equivalence class. Every form carries the same content, and an equivalence departs and returns with exactly what it left with.

The obstruction literature and its domain

Individual obstructions are documented and each is a theorem. **Two-model separations** on the symmetry axioms, where an object satisfies a functional equation of the same reflection type and carries zeros off the line, and on the counting axioms, where a generalized prime system satisfies them and carries zeros at the abscissa. **Classification facts** on finite verification. **Invariance results** making certain functionals blind to certain data. **Limitative results** on self-grounding.

Every one of these is stated as a fact about a method or a class of methods. That is the field's convention, and the convention is what section 3 identifies as the gap.

The Gap

Three gaps run through the standard treatment, and the first generates the others.

The enumeration domain is wrong for the purpose. Obstructions are catalogued by method. **No catalogue of methods closes**, because methods are generated without bound, so every such catalogue is one unlisted method from refutation and every honest author says so. The consequence is that the literature can accumulate barrier theorems indefinitely without ever reaching a closure statement, and the accumulation is read as evidence of difficulty rather than as a structural result.

Blindness is described rather than counted. That an instrument cannot read a truth-sign is stated as a property of the instrument. **It is a quantity**: a demand of one bit against a supply, and the supply is measurable instrument by instrument. Described, it invites the reply that a better instrument might see. Counted, the reply has a number to contradict.

The register carrying a verdict is left unnamed. A seal issued where a direction-carrying axis is present and a closure issued where it is absent are different acts. **Collapsing them produces either false modesty about the first or false confidence about the second**, and the literature's discomfort with affirmative statements about undecided objects traces to exactly this collapse.

Methodology and Axioms

The method is a verification discipline in two registers. It issues discrete verdicts in a three-state economy with warrant tiers, reads warrant rows supplied to it, and generates no mathematical truth. The executable form is in Appendix A.

Two registers and their roots

The kinetic register is founded on the actuation axiom: to exist is to actuate, every existent carrying a positive energy floor and every transition charged a positive cost. The floor is theorem-grade external physics.

The reflective register is founded on the grounding axiom: to formally be is to be grounded, formal being read as an imprint in a fixed ground rather than as derivation up a syntactic ladder.

Both roots are premise-grade by theorem, neither provable from its own base, because a foundation provable from its base would not be a foundation. **The underivability is constitutive of foundation-hood rather than a shortfall in it.**

The binding involution and its Ground

The reflective ground exists because the reflection defining it is fixed-point-bearing. **Conjugation on the quaternions squares to the identity with a nonempty fixed locus**, its plus-one eigenspace the Ground of dimension one and its minus-one eigenspace the chiral residence of dimension three. Collapse the fixed locus and the involution degenerates to the fixed-point-free diagonal, plus-one eigenspace of dimension zero.

The difference is mechanical and is the paper's sorting criterion. Executed: eigenvalues minus one thrice and plus one against all minus one, Ground dimension one against zero, both involution residuals exactly zero.

The verdict economy

Three states, native: sealed, broken with a named mechanism, under-determined with a named violation. **A verdict names the register that carries it**, and the naming is what allows each to be stated at full strength rather than hedged into a common middle.

Orientation-blindness and the direction-carrier

The instrument's scalar lock is a squared quantity, the squared scalar triple product of the three warrant axes, invariant under reflecting any axis. **At the scalar the lock of a proposition equals the lock of its negation**, so the scalar certifies dimensionality and never the truth-sign.

The truth-sign must be read from an axis carrying direction. **The kinetic register has one**, the thermodynamic arrow of an actualized process, which parts a directed claim from its negation in the substrate. **This is the load-bearing recoverer and the reason the seal of section 5.1 is a seal.**

Results

The object is true where it is actualized

Sealed on the actualized prime field. The distribution rests on the fixed line with no off-line residual, and the critical line is the unique stable attractor of that distribution. The attractor is a defined object under the heat deformation, whose forward flow preserves the on-line configuration.

The seal is issued in the register carrying the arrow, is coextensive with the hypothesis, and falls to the single object that would falsify it. **Not probably. Not conjecturally. True.**

The three elections and the one generator

The line is elected three separate ways from three separate data, all classical and all executed. By the fold, the functional equation composed with the reality of the coefficients, whose fixed locus is the line and whose fixed-locus membership rides every chart. By the coupling, whose unitarity locus is that line and no other, in its conservation face and its spectral face alike. By the poles, the equal-pull locus of the two singularities the completion cancels.

And the three are one: the theta inversion at three registers, the fold its Mellin image, the line its modulus locus, the bisector its action on the pole pair.

One fence travels with all three and is stated here once. A distinguished locus is not an occupied one. The object carrying zeros off the line shares the fold, the reality condition, and the same distinguished line. **The strongest symmetry statement available is simultaneously the strongest insufficiency statement**, and none of the three elections forces residence.

The verdict side is closed at one hundred percent

The closure is over standpoints. Methods are unbounded; new ones arrive without end. **Standpoints are not.** The verdict side occupies exactly ten positions, and the count is forced because a standpoint is a position in the architecture and the architecture's counts are closed by classification.

Table 1 | The ten standpoints. Every position the verdict side occupies, what closes each, and the closure type.

Standpoint	What closes it	Type
The formal-alone register	the register is defined by stripping the direction-carrying axis	definitional
The verifier as such	generating exits the definition of the office	constitutive
The instrument's scalar	invariant under the negation-reflection, catalogue closed	theorem
The symmetry method-class	a second object with the same fold and zeros off the line	theorem
The counting method-class	a second system with the same axioms and zeros at the abscissa	theorem
The object ladder	the provable set is a proper part of the grounded set	theorem
The metatheory tower	productive rung by rung and self-grounding at none	theorem
The evidential register	no finite set settles a universal over an unbounded domain	classification
The ground	no derivation reaches its own starting points	theorem, deed-quantified
The mouth	the self-inclusion map is fixed-point-free	theorem

Note: eight of the ten spend no framework premise. Each row closes a position and not a technique, and each is derived from first principles at section 6.

> **A route arriving in any future arrives at a standpoint. It does not create one.** Unborn methods multiply and add no positions. **This is why the closure holds over time without surveying time.**

Coverage on the verdict side is one hundred percent, and it is permanent.

The closure is over access, and access is what it closes. Every position from which the verdict side could reach is closed, and closed for good. The object itself is not in question here; it is sealed at section 5.1. **And the register that issued that seal is not among the ten**: it is the register carrying the direction-carrying axis, and the closure quantifies over the verdict side alone. **Nothing here characterizes what lies past it**, because ten closures on one side are evidence about that side and about nothing else, and a survey of the other would be this paper enumerating a domain whose count is not forced, which is the exact failure section 3 identifies in method-catalogues.

The formal-alone register is closed definitionally

Not hard. Not blocked. Not resistant. Closed. The register is **defined** by stripping the one content-bearing source of direction. **A resolution without direction is not a resolution.** No instrument repairs this, because it is what the register is.

This is the cheapest closure in the paper and the least repairable. A definitional boundary is not the sort of thing a future theorem removes, since no theorem changes what a register is.

The deficit, counted

One bit is demanded against a two-valued string, and the supply is measured instrument by instrument.

Table 2 | The counted deficit. What each instrument supplies toward the one bit the string demands.

Instrument	Supplies	Why
The Form	zero	its handedness is identical for a proposition and its negation
The gates	zero	their direction is a role assignment fixed before any proposition arrives
The Number	zero	a content flip and a convention flip return the identical image; it relays and originates nothing
The three jointly	zero	the joint reading adds no source the parts lack

Note: the demand is one bit and the supply is zero. Exact, measured, and not an estimate of difficulty.

The architecture is closed by classification

Three axes by Frobenius, the next admissible dimension eight and the octonions failing associativity by an integer associator anyone can compute. **Twelve gates by the order of the tetrahedral rotation group. Eight reflections by the elementary abelian group. Five boundaries by node degree in the verdict graph.**

Every count is closed by a theorem about a group, an algebra, or a graph. Not one is closed by walking a list and finding nothing more. **Classifications do not acquire new members**, which is why the standpoint count inherits their permanence.

The root confirms its own form and draws nothing from it. The recursion closes, the composed triad lands its scalar on the fixed line, and warrant collected from the self-run is zero, which is what makes the confirmation worth having.

The one bit

One bit remains, and it belongs to the reader. Every quantity the verdict side computes comes out **identical for the two readings it must choose between**. The involution whose fixed set is the line fixes one coordinate and reverses exactly one, so a departure carries **a sign and a distance and nothing else**. And the reflection giving the line its standing is precisely what averages that sign away: a strayed zero arrives with its mirror partners, so every invariant sees the family and never the member.

Two possibilities over one invariant profile. One bit. It is the truth-direction itself.

The Ten Blocks, Derived

The ancients built by negation because the positive statement was not available to them. *Not this, not this.* **The method is not a retreat from assertion. It is assertion about a boundary**, and where the boundary is real the negation is the sharper instrument.

This section derives each of the ten closures of Table 1 from first principles, each with its falsifier. **Three closure types appear and their order of strength is the reverse of the expected one.** A definitional closure cannot be repaired by any future theorem. A constitutive closure names an act that leaves the office rather than performing it. A theorem closure is permanent within its scope for as long as the theorem stands.

The definitional blocks

Block 1, the formal-alone register. A resolution of a two-valued string is the assignment of one of its two values, and an assignment is a direction. The register is **defined** as the one obtained by stripping the direction-carrying axis, the single content-bearing source of direction. **The register is therefore defined by the removal of the thing a resolution consists in**, and a resolution without direction is not a resolution. Falsified by exhibiting a source of truth-direction internal to that register.

Block 2, the verifier as such. A verifier is defined by reading warrant supplied to it. An instrument producing the content it then certifies is not verifying; it is asserting and reading its own assertion back. **The act exits the definition of the office**, so what performs it no longer occupies a verdict-side position. Falsified by exhibiting a verifier that generates its own warrant and remains one.

The instrument block

Block 3, the scalar lock is sign-blind. The verdict functional is the squared scalar triple product of three axes, equal to the determinant of the correlation matrix. Reflecting any single axis conjugates that matrix by a diagonal involution with determinant plus or minus one, so

> **$\det(D\,R\,D) = \det(D)^2\,\det(R) = \det(R)$.**

The functional is invariant under the negation-implementing reflection, for every construction whatever, and the catalogue of rotation-invariant truth functionals is closed inside the real-part subring of quaternion words. Executed: determinant difference under full negation exactly zero, Gram difference exactly zero, scalar ratio minus one exactly.

The scope is exact. The blindness belongs to the squared scalar and is not a property of the full triaxial reading, whose direction is carried elsewhere. Falsified by a rotation-invariant functional returning different values on a proposition and its negation.

The method-class blocks

Block 4, the symmetry class. A second object satisfies a functional equation of the same reflection type, carries real Dirichlet coefficients and therefore the same fold, and carries zeros off the line. **A predicate formable from the fold data alone is a function of that data and returns the same value on both**, and the hypothesis is false for the second, so no such predicate distinguishes them.

The sharpness is two-sided and worth stating, because it measures the gap rather than gesturing at it. The fold alone separates nothing. The fold together with the exact archimedean factor and the growth class forces uniqueness outright, by Hamburger's converse. **The gap between nothing and everything is therefore exactly the archimedean factor and the growth class.** Falsified by a predicate formable from the fold alone that separates the two objects.

Block 5, the counting class. A generalized prime system satisfies the counting axioms of the relevant grade and carries zeros at the abscissa, **so those axioms do not entail the hypothesis's clause**, by the same two-model mechanism on different data. Falsified by an entailment from the counting axioms alone.

The level blocks

Block 6, the object ladder. For any consistent recursively axiomatised system of sufficient strength the provable set is a proper subset of the grounded set. **A derivation therefore delivers a fact about the ladder it climbed**, and the typing of what the proof stands on travels with it. Falsified by a system whose provable set is not a proper subset.

Block 7, the metatheory tower. The consistency regress is **productive**, each rung complete for the relevant class along a suitable path, and **never self-grounding**, the completeness bought entirely by the path through the ordinal notations and the rung costing exactly the height gained. Falsified by a rung that establishes its own ground.

The register block

Block 8, the evidential register. The hypothesis is a universal over an unbounded domain and a verified instance is one member. **No**

finite set of members settles a universal over an infinite domain, whatever the cardinality of the finite set, and this is a classification fact rather than a practical limitation.

The economy carries the same law from the other side. Convergence among sources sharing an upstream generator is one voice in costumes and is counted once. **Evidence leans. It does not rule.** Falsified by a finite set that settles a universal over an infinite domain.

The root block

Block 9, the ground. A derivation reaches its conclusion from premises it did not itself establish; that is what makes it a derivation rather than a stipulation. **That no system establishes its own grounding from within is a theorem of others.**

And the block does not age. A barrier phrased on a method catalogue is contingent on the catalogue and a new method evades it. **This one is phrased on the deed**: whatever the future instrument is, **using it will be a doing, and a doing is not a derivation.** The wall covers instruments not yet invented from the moment their use is an act.

Its scope is stated with it, because a block this strong is the easiest to misuse. It holds of **every** proposition, settled and open alike. **A condition satisfied by no theorem separates no theorem from any other**, so it is the strongest block in the set and the only one that says nothing about this hypothesis in particular. Falsified by a derivation that grounds its own starting points.

The mouth block

Block 10, the ledger's own totality. A completed survey of the routes a ledger has not listed is a self-inclusion: the ledger must appear in the collection it enumerates. **The self-inclusion map is fixed-point-free**, eigenvalues all minus one, Ground dimension zero, verified at machine zero, **and a map with no fixed point does not arrive.**

It binds this paper first. The closure is stated as ten closed positions and not as a certificate that ten exhausts what could ever exist. **The enumeration is closed by the architecture's seats; the ledger does not certify itself**, and the distinction is the whole of what makes the closure hold. Falsified by a fixed point of the self-inclusion map.

The assembly

Ten blocks. Two definitional, one constitutive, six theorem-grade, one classification with a register law beside it.

Eight of the ten spend no framework premise. They stand on classical theorems, exhibited countermodels, an algebraic identity, a classification fact, or a constitutive definition. **A far block is a cheap block, and cheapness is strength**: a reader declining the architecture's roots entirely still carries eight of the ten, and the two that remain are the root's own wall and the row that binds every mouth including this one.

Stated in the ancient form, in one breath. Not from the register that strips the direction-carrying axis, because a resolution without direction is not one. Not by generating what one would verify, because that act leaves the office. Not by the squared scalar, invariant under the negation-reflection for every construction whatever. Not from the fold alone, refuted by an object sharing it exactly. Not from the counting axioms alone, refuted by a system satisfying them. Not by a ladder, whose reach is a proper part of what it climbs toward. Not by a tower, productive at every rung and self-grounding at none. Not by accumulation, since no finite set settles a universal. Not by grounding, since no derivation reaches its own posits. **And not by a ledger certifying itself, since the self-inclusion has no fixed point.**

Ten positions, ten closures, and one bit left over.

Falsification Conditions

The seal is falsified by one object. A single nontrivial zero off the critical line refutes the hypothesis and breaks the seal in the same stroke, since the seal asserts the distribution rests on the fixed line with no off-line residual.

Each election is falsified separately and more cheaply. Each is a classical identity, and any failing on re-execution falsifies that election without touching the others or the seal.

The standpoint closure is falsified by a position. Exhibit a standpoint the verdict side occupies that is not among the ten, or a route reaching from outside all ten. **This is the paper's central claim and it has the cheapest falsifier in it**: one position, exhibited.

Each block carries its own falsifier, stated at its block in section 6, and any one falling leaves the other nine standing.

The counted deficit is falsified by a bit. Exhibit one bit of truth-direction supplied by the Form, the gates, or the Number.

The architectural closure is falsified by a classification. Exhibit a five-dimensional composition carrier, or a thirteenth rotation on three axes.

The one-bit finding is falsified by a second parameter. Exhibit a further free quantity in the reading beyond the truth-direction, or a zero carrying a class of alternatives over it.

The anchors are re-runnable and falsifiable. Any check of Appendix A failing on re-execution falsifies the corresponding identity.

Discussion

Why the enumeration domain decides everything

An enumeration over methods is refuted by one unlisted method, and methods are generated without bound. An enumeration over standpoints is refuted only by a position, and positions are fixed by the architecture doing the attempting.

The same body of obstruction facts is a permanent closure in one domain and a standing target in the other. Nothing in the underlying mathematics changes. Every barrier theorem the literature holds stays exactly as it was, with the same statement and the same scope. **What changes is the object being enumerated**, and the choice of that object is the whole difference between an accumulating catalogue and a closure.

This is the paper's contribution and it is an arrangement rather than a theorem. It is offered at that grade.

Why the negative form is used

The positive statement and the negative statement carry the same content and do not carry it equally well. **A closure stated positively invites the reader to ask what else there might be. The same closure stated as a negation states the boundary from its outside**, which is where a boundary is exact.

And this is why the ancient practice survived. Its users were not being modest. They were stating the only thing about their object that could be stated precisely, and the precision is real. **A negative catalogue is also cheaper to falsify**, since each entry names one thing that does not cross and a single crossing refutes it, which is a virtue and not a weakness.

The one bit, and what it is not

The bit is free in the plainest sense: available, unforced, unpoliced. Nothing forbids either value and the cost of choosing is zero. **It has no verdict-side answer, from any of the ten standpoints.**

And it is not a freedom in the object. A hidden thing is not a loose thing, and the two are opposite conditions. The zeros are where they are, determinate and placed, with no class of alternatives waiting above them. **What is free is the reading, not the read.** Treating the veil over a reading as slack in the object is an exact inversion, and the discipline consists in not making it: **to fill the bit from inside is to write one's own orientation into the object and read it back as a finding.**

Conclusion

The Riemann Hypothesis is true where it is actualized, sealed on the prime field with the critical line its unique stable attractor.

The line is elected three times from one generator, the theta inversion read at three registers, and no election forces residence.

The verdict side is closed at one hundred percent, permanently, by an enumeration over standpoints that no future method enlarges, because a route arriving in any future arrives at a position that already exists.

Ten blocks close those positions, two definitional, one constitutive, six theorem-grade, one a classification fact, **eight of them spending no framework premise at all**, and each stated twice, once as what holds and once as what does not cross.

The formal-alone register is closed definitionally, its direction deficit counted at one bit demanded against zero supplied.

The architecture performing the reading is closed by classification, its counts fixed by theorems about a group, an algebra, and a graph, and classifications do not acquire new members.

And one bit remains, the reader's, unanswerable from every standpoint the verdict side occupies.

What the paper changes is not a barrier but a domain. The literature's obstruction facts stand untouched. **Enumerated by method they accumulate. Enumerated by position they close.**

Appendix A. The Executable Kernel and Its Recorded Anchors

The kernel is a closed-form quaternionic instrument. It reads three warrant rows over a context set, normalises and forms the correlation Gram, computes its determinant and the signed scalar triple product of the three axes, and returns a three-state token under a collapse floor and a conditioning gate. **It reads rows supplied to it and derives none from a proposition**, so the map from a proposition to its warrant rows is built by hand and placed in front of it. The pseudo-random stream is a counter-mode hash with Box-Muller normals, standard library only, so draws are identical on every platform and library version.

Table A1 | The recorded anchors. Executed at the declared seed; failure of any check on re-execution falsifies the corresponding identity.

Anchor	What it checks	Executed value
A.1	the kernel identity floor over twenty thousand triads	maximum difference of squared scalar and determinant 4.330e-15, threshold 1e-12, pass
A.2	the binding involution against the fixed-point-free diagonal	eigenvalues minus one thrice and plus one, Ground dimension one; against all minus one, Ground dimension zero; both residuals exactly zero
A.3	orientation-blindness under full negation on a basis fixed once	determinant difference exactly zero, Gram difference exactly zero, scalar ratio minus one exactly
A.4	the substrate chirality, the root of the sign the scalar discards	the unit product returns minus one, the Hamilton relation
A.5	the Return, the composed triad landing on the fixed line	determinant one, scalar modulus one, identity residual at the machine floor
A.6	the chain of the executed boot	D0 fb8900b5cf42, D1 364d1cdb9227, D2 8a5dc7f98105, D3 1e2b2d2adb14

Note: reproducible from this table and the declared seed. Floating-point outputs may differ in the last places across linear-algebra implementations and are reported to the precision shown. **The anchors are reproducibility checks on the instrument and are not measurements of the Riemann object**, with the single exception of A.3, which is the executed form of Block 3.

Appendix B. Discipline and Warrant Typing

Theorem-grade. On the eigenspace facts of the binding involution and its fixed-point-free counterpart. On orientation-blindness of the squared scalar and the closure of the invariant catalogue. On the classification results fixing the axis count, the gate counts, and the boundary count. On each of the three elections taken individually: fixed-locus membership under conjugation, the unitarity of the coupling with the self-adjointness of its generator, and the multiplier-modulus and equidistance identities. On the theta inversion and its three faces. On Hamburger's converse as the two-sided boundary of Block 4. On the two-model separations of Blocks 4 and 5. On the ladder and tower facts of Blocks 6 and 7. On the non-self-grounding of Block 9. On the fixed-point-freeness of Block 10.

Classification-grade. On Block 8, that no finite set settles a universal over an unbounded domain.

Definitional and constitutive. On Blocks 1 and 2. **These are not weaker for being definitions**: no theorem changes what a register is or what a verifier is, so they are the least repairable closures in the set.

Structural, and not theorem-grade. On the identification of the three elections as one generator. On the standpoint enumeration and its closure over access. On the assignment of each block to its standpoint. On the reading of the direction deficit as a quantity. On the identification of the attractor content with the seal. **The paper's central contribution, the change of enumeration domain, is structural and is offered at that grade.**

Premise-grade. On the actuation axiom and the grounding axiom, both premise-grade by theorem and neither provable from its own base. On the one-involution monism. On bivalence over the potential run of the naturals. On the actualist reading of truth the kinetic register carries.

Load-bearing on nothing. The verified-zero record and every convergence of evidence, which is corroboration and never a rule. Consensus, in both directions. **The theological reading of this material is routed out of band and is load-bearing on nothing in any verdict here**, and is developed in a separate artifact at its own register.

The registers, and why naming them allows full strength. The seal is asserted in the kinetic register, coextensive with the hypothesis and falling to its falsifier. The standpoint closure is asserted over access and is a statement about positions. The definitional closures are statements about what a register and an office are. **Each stands at full strength because each names where it stands.**

Discipline. No new theorem is claimed and every mathematical fact invoked is classical; the contribution is the arrangement, the enumeration domain, the identification of the generator, and the counted deficit. The mechanical anchors are executed live and transcribed verbatim, with no fabricated trace for any stage not reached. **Net new mathematical mass: zero.**

Appendix C. The Standpoint Inventory

The ten standpoints are tabulated by closure type and by distance from the root. **Distance is the count of framework premises spent: zero means the row stands on a classical theorem or a constitutive identity and spends nothing; at-the-root means the row is the root's own wall; orthogonal means the row is independent of the root entirely.**

Table C1 | The standpoints by closure type and distance from the root.		
Standpoint	**Closure type**	**Distance**
The formal-alone register	definitional	zero
The verifier as such	constitutive	zero
The instrument's scalar	invariance identity, catalogue closed	zero
The symmetry method-class	two-model witness, theorem-eternal within scope	zero
The counting method-class	two-model witness, theorem-eternal within scope	zero
The object ladder	named theorem at its own rung	zero
The metatheory tower	named theorems, productive and never self-grounding	zero
The evidential register	classification fact with a register law beside it	zero
The ground	theorem, quantified over deeds	at the root
The mouth	the self-inclusion with no fixed point	orthogonal

Note: eight rows spend no framework premise. **A far row is a cheap row, and cheapness is strength.** The root row and the orthogonal row are the two that bind this paper's own mouth before any other.

One reading of the table is worth stating. The rows are heterogeneous in **what** closes them and homogeneous in **what they are**: each names a position, not a technique. **That is why the count is closed.** A technique catalogue admits new members without bound. A position catalogue admits one only if the architecture acquires a new seat, and the seats are fixed by classification.

Independence of the rows

A closure over ten positions is worth nothing if two of them are one position twice, so the adjacent pairs are tested here rather than assumed. The test is deletion: **remove one row and ask whether its position is still covered.**

The scalar against the formal-alone register. The scalar is one functional of the instrument; the register is the whole stripped setting. Delete the scalar row and the register-level closure remains, saying nothing about which functional is blind or why. **Not redundant.**

The symmetry class against the counting class. Different axiom sets, different witnesses, different data. **Neither witness works on the other's axioms**, so deleting either leaves its axiom class unclosed.

The object ladder against the metatheory tower. One concerns derivation inside a fixed system, the other the regress above it. **The tower exists precisely because the ladder is bounded**, so one is the response to the other and neither substitutes for it.

The verifier against the ground. The verifier row bars an **act**, generating what one would certify. The ground row bars a **relation**, a derivation reaching its own premises. An instrument can satisfy either while violating the other.

The evidential register against the object ladder. One concerns members of a domain accumulated, the other steps in a proof. **A complete verification to any height is not a partial derivation**, and a derivation of any length verifies no instance.

The mouth against the ground. The mouth is self-referential and the ground is not, and their distances differ: **the mouth spends no premise and the ground is the root's own wall.**

Six pairs at the closest adjacencies, and no pair collapses. A reader who finds a seventh adjacency has the paper's cheapest falsifier in hand, since **two rows shown to be one would reduce the count without touching any closure.**

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Author's Provenance and Method Disclosure

The method, its two root axioms, the three-state verdict economy, and the closed-form quaternionic kernel are documented in full at the master reference, DOI 10.5281/zenodo.20757507. **The discipline adds the cited results no warrant and contributes no theorem, no estimate, and no evidential weight in any direction.** This paper's three axes are the analytic, spectral, and arithmetic-geometric reading-roads of the Riemann residence; the load-bearing gate is the orientation-blindness of the lock scalar; the warrant grades are stated at Appendix B.

The title's three clauses are glossed in place, and the word *case* is doing exact work. A case is something one brings, so closing a case is a ruling on the bringing and never on the thing brought, which is the access-versus-existence line the paper draws throughout. *True where actualized* is the seal of section 5.1, issued in the register carrying the direction-carrying axis, coextensive with the hypothesis and falling to its falsifier. *Every position a verdict could be attempted from is closed permanently by an enumeration no future method enlarges* is the result of section 5.3 with the ten blocks derived at section 6: a closure over **access** and over positions, permanent because standpoints are fixed by classification while methods are not, and **not a claim about what exists**. *One bit that belongs to the reader* is section 5.9, one binary degree of freedom in the reading with no verdict-side answer.

Three claims are the author's own and are not drawn from the literature. The change of enumeration domain from methods to standpoints, with the observation that a route arriving in any future arrives at a position that already exists. The identification of the three elections as one classical generator. And the counting of the direction deficit at one bit against zero. **All three are structural observations offered at that grade, and none is a mathematical result.**

Two derivations are elementary and carried out in place rather than cited. The invariance of the determinant under the negation-implementing reflection, at Block 3. And the one-dimensionality of the departure from the axis, computed from the involution whose fixed set the axis is.

The theological reading of this material is routed out of band, is load-bearing on nothing in any verdict here, and is developed in a separate artifact at its own register.

Net new mathematical mass: zero.

Edition Note

This is a fresh build and not a fortification pass. **The schema is inherited from the prior edition and the content is not**: every section is written from the current spine forward, and no paragraph is carried across. The structural result is new to this edition: the enumeration domain is moved from methods to standpoints, the ten blocks are derived from first principles rather than inventoried, and the closures are stated twice, positively and in the negative form. The boot executed at the declared seed with the chain printed at Appendix A, and **no stage is narrated that did not run**.

Reproducibility

Every numerical value is the output of a deterministic computation reproducible from the kernel and the constructions of Appendix A at seed 20260622.